

The Prime Prediction Budget

How far structure gets you toward the next prime, and the wall it hits

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What this is. A synthesis — an honest ledger, not a discovery. It stacks *every* usable regularity for locating a prime, sorts what actually pays from what is the same fact in disguise, and reads off the information-theoretic ceiling that caps the whole game. **Every ingredient is classical** — the prime number theorem, the Hardy–Littlewood wheel, the Cramér model, the parity barrier (Sarnak’s Möbius randomness). What this note adds is the *assembly*: one calibrated budget, anchored to the Factor Skyline entropy numbers, that says precisely how far you can get and where you provably stop. Tags: **[deriv]** from known theory · **[emp]** measured here · **[interp]** synthesis. Companions: [Switchback Law](#), [Prime-Gap Memory and the Differencing Trap](#), [The Prime-Triangle Angle](#), [Seven Sisters](#).

1. The question, posed so it has an answer

“Predict the next prime” is the wrong target — the exact prime is not predictable, and the whole point of this note is *why*. The right target is bounded: **from a known prime p , how few candidates must you test to be, say, 90% sure you have caught the next prime — and what stops you from doing better?** That question has a clean, quantitative answer, and the answer is almost entirely *one* structure.

2. The budget, in three tiers

Tier 1 — load-bearing (this is essentially all of it).

- **Base rate (PNT).** Near x , an integer is prime with probability $\sim 1/\ln x$; the expected gap is $\ln x$. This sets the scale — look near $p + \ln p$.
- **The wheel (admissibility).** Only positions coprime to the small primes can be prime: the mod-30 wheel leaves 8/30 of integers open, mod-210 leaves 48/210 — killing 73–77% of all positions *for free*, with no test. This is the singular series / Hardy–Littlewood structure, and in Factor-Skyline entropy terms it is worth ≈ 1.70 **bits** of the ≈ 2.48 -bit local uncertainty — about 68% of everything knowable.

Tier 2 — real, but not extra bits (the wheel wearing hats).

Each of these is a genuine, measured regularity; none of them *adds* predictive power beyond Tier 1, because each is a face of the same wheel:

regularity	what it is	(established in)
jumping champions 6, 30, 210	the wheel’s favoured gaps	classical (Odlyzko–Rubinstein–Wolf)
gap anti-correlation -0.05 switchback sign-law (reversal $63 \rightarrow 82\%$)	big gap $\rightarrow$ smaller next runs die faster than a coin	wheel’s consecutive-gap memory Switchback Law
mod-6 / LOS bias	consecutive gaps avoid a shared residue	Lemke Oliver–Soundararajan
Seven Sisters ($2p + k$, $\sim 80\%$)	the wheel at the doubled scale	Seven Sisters

Tier 3 — the wall.

- **The escape / parity barrier.** After the wheel, the residual — *which* open slot is actually prime — is genuinely random: the Möbius/Cramér noise, the parity barrier. In the FS entropy budget this irreducible core is only about 0.26 **bits** (its peak). It is not hard to compute; it is *impossible* to predict. This is the wall, and it is what “the information theory of primes” means.

3. The number: ~7–8 candidates for 90%

Put Tier 1 to work — from p , walk *up the wheel-open slots* and test them in order — and measure how many you need. Over the primes near 10^6 :

open slots tested	1	2	3	4	5	6	7	8
P(next prime caught), mod-210	30%	52%	67%	77%	85%	90%	93%	95%
P(next prime caught), mod-30	25%	45%	60%	71%	79%	85%	89%	92%

[emp] So: **the next prime is within the next ~7 (mod-210) or ~8 (mod-30) wheel-open slots about 90% of the time** — a mean of ~3.2–3.7 tests. That 90% is bought *entirely* by base rate + wheel.

4. Why the rest of the list doesn’t help — and why that’s the point

It is tempting to think the Tier-2 biases should tighten the list. They do not, and the reason is exact: to land on the k -th open slot, every earlier open slot must be composite, so $P(\text{next prime} = k\text{-th slot})$ **strictly decreases in k** — *walking up in order is already the Bayes-optimal search*. Reweighting the candidates by gap-frequency never moves a different slot to the front (0% of cases), and the full recent-history “regime” lever — given every advantage — shifts the mean rank by +0.0007. The biases are real physics of the wheel; they are simply *already spent* the moment you respect admissibility. Everything past the wheel is the 0.26-bit escape, and the escape is, by theorem-strength heuristic, unbeatable.

So the honest ceiling is not “predict the prime.” It is: **pin it to ~7–8 candidates with 90% confidence, and know that no cleverness — no bias, no history, no lens — does better, because the remainder is the parity barrier**. The wheel alone already touches that ceiling; the entire rest of this collection is the wheel proving, six different ways, that it is the whole story.

5. Honest ledger

Nothing here is new mathematics. The wheel is Hardy–Littlewood; the ceiling is Cramér and Sarnak; “primes are unpredictable past the density” is the working assumption of the whole field. The contribution of this note is **calibration and assembly** — one page that separates the load-bearing structure from its many disguises, attaches the FS entropy numbers (1.70 bits captured, ~ 0.26 irreducible), and states the reachable target as a number (~ 8 candidates for 90%) rather than a hope. Its value is knowing *exactly* how far to trust structure, and exactly where to stop trying.

Appendix. Reproduction

```
import bisect
from math import gcd
def sieve(n):
    b=bytearray([1])*(n+1); b[0]=b[1]=0; i=2
    while i*i<=n:
        if b[i]: b[i*i::i]=bytearray(len(b[i*i::i]))
        i+=1
```

```

    return b
B=sieve(2_000_000)
primes=[i for i in range(1_000_000,1_500_000) if B[i]]
def slots_for_90(wm):
    ranks=[]
    for p in primes:
        k=0; n=p
        while True:
            n+=1
            if gcd(n,wm)!=1: continue      # skip non-wheel-open positions
            k+=1
            if B[n]: ranks.append(k); break
    ranks.sort()
    for k in range(1,20):
        if sum(1 for r in ranks if r<=k)/len(ranks) >= 0.90: return k
print('mod-30:', slots_for_90(30), ' mod-210:', slots_for_90(210))    # 8, 7

```

References

- Prime number theorem; G. H. Hardy & J. E. Littlewood, *Partitio numerorum III* (1923) — the singular series (the wheel).
- A. M. Odlyzko, M. Rubinstein, M. Wolf, *Jumping champions* (Exp. Math., 1999).
- R. J. Lemke Oliver & K. Soundararajan, *Unexpected biases in consecutive primes* (PNAS, 2016).
- H. Cramér, *On the order of magnitude of the difference between consecutive prime numbers* (1936); P. Sarnak, *Möbius randomness and dynamics* (2011) — the escape / parity barrier.
- A. Proxmire, *Factor Skyline* (this collection) — the entropy budget ($H \approx 2.48$ bits, template ≈ 1.70 , escape ≈ 0.26); *Switchback Law, Differencing Trap, Seven Sisters* — the Tier-2 regularities and the Bayes-optimality of walking the wheel.